

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(ggplot2)
library(agricolae)
library(lubridate)
library(here)

## here() starts at /home/guest/EDE_Fall2025

library(knitr)
library(dplyr)

NTL_ChemPhys <- read.csv(
  here("Data", "Raw", "NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE)

NTL_ChemPhys$sampldate <- as.Date(NTL_ChemPhys$sampldate, format = "%m/%d/%y")

#2

mytheme <- theme_classic(base_size = 14) +
  theme(plot.background = element_rect(fill = "pink", color = NA),
        axis.text = element_text(color = "black"),
        axis.title = element_text(face = "bold", color = "purple"),
        legend.position = "bottom")
theme_set(mytheme)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The mean lake temperature recorded during July does not change with depth across all lakes. Ha: The mean lake temperature recorded during July changes with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
## 'geom_smooth()' using formula = 'y ~ x'
```

6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure suggests that the temperature generally decreases with depth across all Lakes.

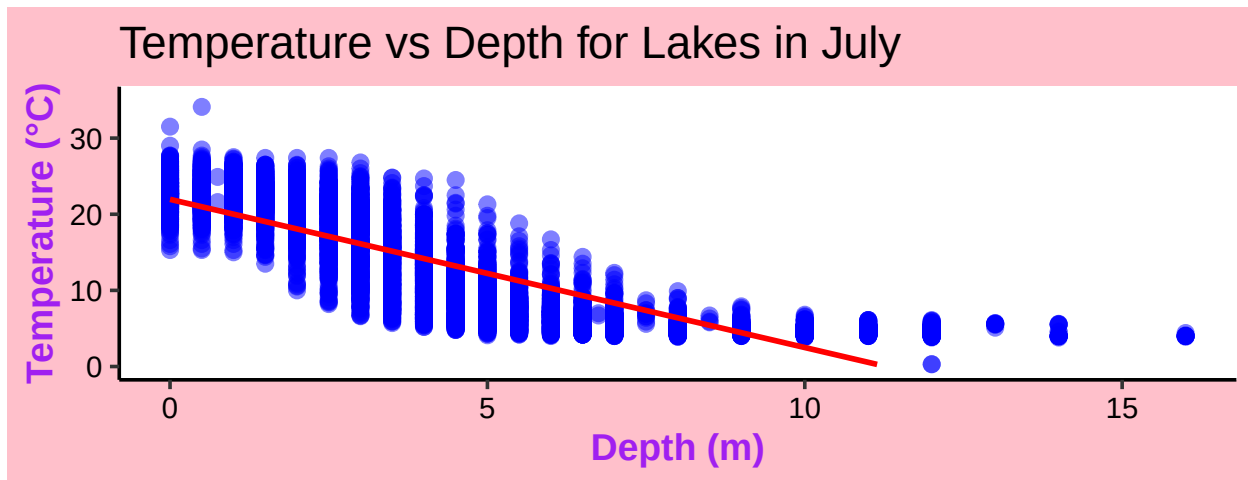


Figure 1: Temperature Vs. Depth for Lakes in July

7. Perform a linear regression to test the relationship and display the results.

```
#7

lm <- lm(temperature_C ~ depth, data = NTL_ChemPhys_Filter)
summary(lm)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_ChemPhys_Filter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173  -3.0192   0.0633   2.9365  13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth        -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The results of the linear model and the scatterplot indicate that there is a negative relationship between temperature and depth (i.e. temperature decreases with depth). The R-squared value (0.744), indicates that approximately 74.4% of the variability in temperature is

explained by changes in depth. This finding is based on 392 degrees of freedom. The p-value, $< 2.2\text{e-}16$, indicates that the model is statistically significant and confirms that depth has a significant effect on temperature. The slope of the model indicates that the temperature is predicted to change -1.913 degrees for every 1 m of depth.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9

```
AIC_model <- lm(temperature_C ~ year4 + daynum + depth,
               data = NTL_ChemPhys_Filter)
AIC(AIC_model)
```

```
## [1] 53674.39
```

```
AIC_1 <- lm(temperature_C ~ year4, data = NTL_ChemPhys_Filter)
AIC_2 <- lm(temperature_C ~ daynum, data = NTL_ChemPhys_Filter)
AIC_3 <- lm(temperature_C ~ depth, data = NTL_ChemPhys_Filter)
AIC_4 <- lm(temperature_C ~ year4 + daynum, data = NTL_ChemPhys_Filter)
AIC_5 <- lm(temperature_C ~ year4 + depth, data = NTL_ChemPhys_Filter)
AIC_6 <- lm(temperature_C ~ daynum + depth, data = NTL_ChemPhys_Filter)
AIC_7 <- lm(temperature_C ~ year4 + daynum + depth, data = NTL_ChemPhys_Filter)

AIC(AIC_1, AIC_2, AIC_3, AIC_4, AIC_5, AIC_6, AIC_7)
```

```
##      df      AIC
## AIC_1  3 66819.14
## AIC_2  3 66796.54
## AIC_3  3 53762.12
## AIC_4  4 66798.34
## AIC_5  4 53756.97
## AIC_6  4 53679.36
## AIC_7  5 53674.39
```

#10

```
Temp_predictor <- lm(temperature_C ~ year4 + daynum + depth,
                   data = NTL_ChemPhys_Filter)
summary(Temp_predictor)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_ChemPhys_Filter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: Based on the above AIC, the final set of explanatory variables suggested to predict temperature in the multiple regression are year4, daynum, and depth (all of the variables tested). The r-squared of the model, 0.7802, indicates that approximately 78.02% of the variance between variables is explained. This is higher than the previous model, 74.4%, indicating an improvement over the model using only depth as the explanatory variable.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12

#Anova model

NTL_ChemPhys$sampldate <- as.Date(as.character(NTL_ChemPhys$sampldate),
                                format = "%Y-%m-%d")
NTL_ChemPhys$month <- month(NTL_ChemPhys$sampldate)
July_temps <- subset(NTL_ChemPhys, month == 7)
anova_july_temps <- aov(temperature_C ~ lakename, data = July_temps)
summary(anova_july_temps)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
```

```
## lakename      8 21642 2705.2      50 <2e-16 ***
## Residuals    9719 525813    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 1116 observations deleted due to missingness
```

```
#Linear model
```

```
LM_july_temps <- lm(temperature_C ~ lakename, data = July_temps)
summary(July_temps)
```

```
##      lakeid      lakename      year4      daynum
## R      :3187  Peter Lake  :3187  Min.   :1984  Min.   :182.0
## L      :2975  Paul Lake   :2975  1st Qu.:1991  1st Qu.:190.0
## T      :1711  Tuesday Lake:1711  Median :1997  Median :197.0
## W      :1130  West Long Lake:1130  Mean    :1999  Mean    :197.5
## E      :1066  East Long Lake:1066  3rd Qu.:2006  3rd Qu.:205.0
## M      : 358  Crampton Lake : 358  Max.    :2016  Max.    :213.0
## (Other): 417  (Other)      : 417
##      sampledate      depth      temperature_C      dissolvedOxygen
## Min.   :1984-07-01  Min.    : 0.000  Min.    : 0.30  Min.    : 0.000
## 1st Qu.:1991-07-29  1st Qu.: 1.500  1st Qu.: 5.50  1st Qu.: 0.300
## Median :1997-07-31  Median : 4.000  Median :10.10  Median : 5.600
## Mean    :1999-04-20  Mean    : 4.343  Mean    :12.72  Mean    : 4.939
## 3rd Qu.:2006-07-10  3rd Qu.: 6.500  3rd Qu.:20.80  3rd Qu.: 8.100
## Max.    :2016-07-27  Max.    :16.000  Max.    :34.10  Max.    :703.000
##                                     NA's    :1116  NA's    :1241
##      irradianceWater      irradianceDeck      comments
## Min.   : -0.337  Min.    : 2.5  DO Probe bad - Doesn't go to zero: 0
## 1st Qu.: 14.520  1st Qu.:380.0  DO taken with Jones Lab Meter    : 0
## Median : 67.000  Median :795.0  NA's                             :10844
## Mean    :211.439  Mean    :734.9
## 3rd Qu.:270.000  3rd Qu.:1038.0
## Max.    :5323.000  Max.    :3032.0
## NA's    :3902      NA's    :4223
##      month
## Min.   :7
## 1st Qu.:7
## Median :7
## Mean    :7
## 3rd Qu.:7
## Max.    :7
##
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: The f-value from the anova, which measures the ratio of variance amongst variables in the model, is 1.098. This value is very close to 1, which indicates that there is not a large difference in the means between variables. Therefore, the mean temperatures do not differ significantly among lakes.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
## 'geom_smooth()' using formula = 'y ~ x'
```

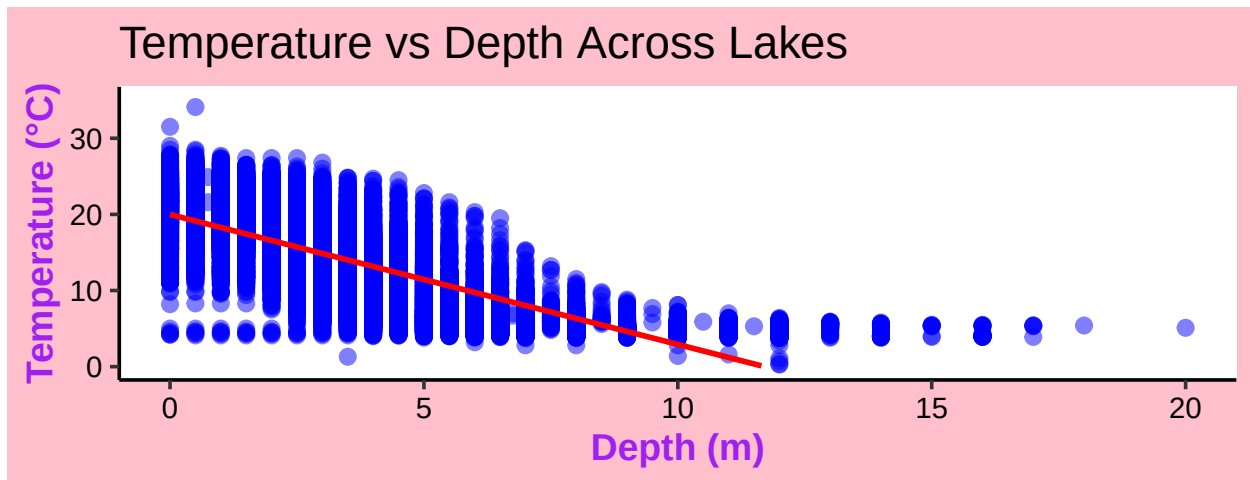


Figure 2: Temperature Vs. Depth Across Lakes

15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
```

```
tukey_test <- TukeyHSD(anova_july_temps)
tukey_test
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = July_temps)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000

## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: According to the results from the Tukey's HSD test, Central Long, Crampton Lake, East Long Lake, Hummingbird Lake, Tuesday Lake, and West Long Lake all have different mean temperatures, statistically speaking, than Peter Lake. This is because the p-adj value is less than or equal to 0.05 for all of those lakes compared to Peter Lake. These results also indicate that Ward Lake and Paul Lake have statistically similar mean temperatures as Peter Lake because all of the p-adj values are greater than 0.05. It appears that none of the lakes are statistically distinct from all of the other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: A t-test would be another useful test to explore if you were just looking at Peter and Paul Lake. This would give you t-value, indicating the variability, and the p-value, indicating the statistical significance.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```
NTL_ChemPhys$month <- month(NTL_ChemPhys$sampldate)

July_Crampton_Ward <- subset(NTL_ChemPhys, month == 7 &
                             lakename %in% c("Crampton Lake", "Ward Lake"))

July_Crampton_Ward$lakename <- droplevels(July_Crampton_Ward$lakename)

t_test <- t.test(temperature_C ~ lakename, data = July_Crampton_Ward)
t_test
```



```
##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862
```

Answer: The results of this t-test indicate that there is no statistical significance between the lake temperatures in July. This is demonstrated by the p value being greater than 0.05 (0.2649). Additionally, the t-value is 1.1181, which is very close to 1, indicating that there is no statistical significant difference between variables. The mean temperatures are not equal - the mean for Ward Lake is slightly lower than Crampton Lake. This matches the answer to part 16, which indicated Ward and Crampton Lake did not have a statistically distance temperature difference (with a p-value of 0.9714459).